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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » **The Joy of Computing using Python (course)**

Announcements (announcements) **About the Course** (https://swayam.gov.in/nd1_noc20_cs35/preview)

Ask a Question (forum) Progress (student/home) Mentor (student/mentor)

Unit 3 - Week 1

Course outline

How does an NPTEL online course work?

Week 0

Week 1

- ☒ Introduction to Programming (unit? unit=1&lesson=2)
- ☐ Why Programming? (unit? unit=1&lesson=3)
- ☐ Programming for Everybody (unit? unit=1&lesson=4)
- ☐ Any Prerequisites? (unit? unit=1&lesson=5)
- ☐ Where to start? (unit? unit=1&lesson=6)
- ☐ Why do we have so many languages?

Assignment 1

The due date for submitting this assignment has passed. **Due on 2020-02-12, 23:59 IST.**
As per our records you have not submitted this assignment.

- 1) Recall the programming using scratch. Which of the following set of instructions will not end up getting the sprite (the cat) back at the initial state (both in terms of movement and the angle)? **1 point**

- ☐
- ☐
- ☐

(unit?
unit=1&lesson=7)

- ☒ How to go about programming?
(unit?
unit=1&lesson=8)

- ☒ Why to learn programming?
(unit?
unit=1&lesson=9)

- ☐ What is programming?
(unit?
unit=1&lesson=10)

- ☐ How to give instructions?
(unit?
unit=1&lesson=11)

- ☐ Introduction to Scratch (unit?
unit=1&lesson=12)

- ☐ Introduction to Loops (unit?
unit=1&lesson=13)

- ☐ More about Loops (unit?
unit=1&lesson=14)

- ☒ Solution to Looping Problem (unit?
unit=1&lesson=15)

- ☐ Scratch : Animation 1
(unit?
unit=1&lesson=16)

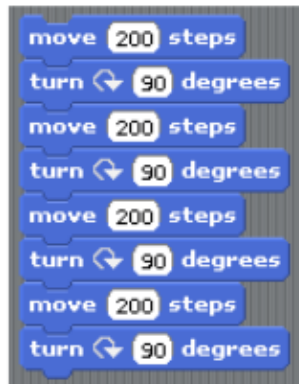
- ☐ Scratch : Animation 2
(unit?
unit=1&lesson=17)

- ☐ Scratch : Animation 3
(unit?
unit=1&lesson=18)

- ☐ More on Scratch
(unit?
unit=1&lesson=19)

- ☐ Quiz : **Assignment 1 (assessment? name=255)**

- ☐ Week 1 Feedback (unit?
unit=1&lesson=260)



No, the answer is incorrect.
Score: 0

Accepted Answers:



2) What is the output of the following

1 point



- ☐ 3
- ☐ 33
- ☐ 30
- ☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

33

3) What is the output of the following

1 point



- ☐ 0
- ☐ 10
- ☐ -10
- ☐ None of the above

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

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No, the answer is incorrect.

Score: 0

Accepted Answers:

-10

4) When we double click the following instructions, the sprite (cat)

1 point



- ☐ keeps moving forward
- ☐ moves forward in steps of 10
- ☐ moves backward 10 steps and then come back to its original position
- ☐ moves forward 10 steps and then come back to its original position

No, the answer is incorrect.

Score: 0

Accepted Answers:

moves forward 10 steps and then come back to its original position

5) Assuming our sprite to be an aeroplane now, the following instructions represent an aeroplane 1 point



- ☐ moving forward
- ☐ falling
- ☐ moving backward
- ☐ making circular motions

No, the answer is incorrect.

Score: 0

Accepted Answers:

making circular motions

6) Consider 2 blocks of instructions shown below for an aeroplane sprite. Choose the correct option from the following

1 point



- ☐ First block represents landing while the second represents takeoff

- ☐ First block represents takeoff while the second represents landing
- ☐ Both the blocks represent takeoff
- ☐ Both the blocks represent landing

No, the answer is incorrect.

Score: 0

Accepted Answers:

First block represents takeoff while the second represents landing

7) Which of the following represents the code block for a jumping baseball which initially jumps to **1 point** a height of 200 and then in every subsequent iteration reaches 3/4th of the height in the previous iteration till it comes to rest.

☐

```

turn 90 degrees
set height to 200
repeat until height = 0
  turn 180 degrees
  move height steps
  wait 1 secs
  turn 180 degrees
  move height steps
  wait 1 secs
  
```

☐

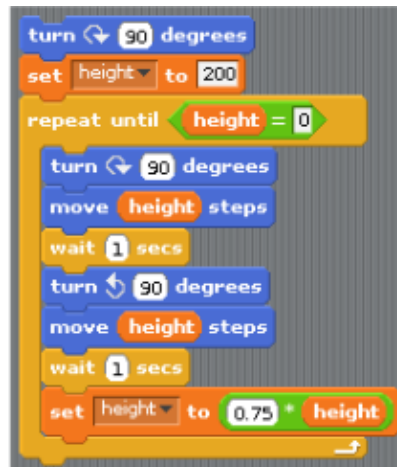
```

turn 90 degrees
set height to 200
repeat until height = 0
  turn 180 degrees
  move height steps
  wait 0.1 secs
  turn 180 degrees
  move height steps
  wait 0.1 secs
  set height to 0.75 * height
  
```

☐

```

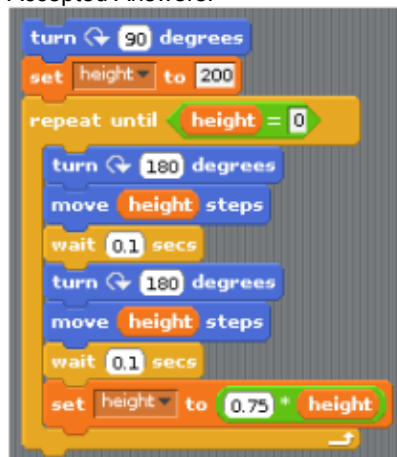
turn 90 degrees
set height to 200
repeat until height = 0
  turn 180 degrees
  move height steps
  wait 0.1 secs
  turn 180 degrees
  move height steps
  wait 0.1 secs
  height = 0.75 * height
  
```



No, the answer is incorrect.

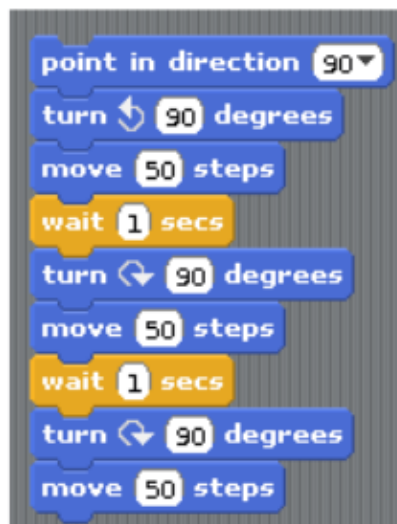
Score: 0

Accepted Answers:



8) Imagine a ghost sprite. What does the following block of instructions represent

1 point



- ☐ Ghost going vertically up and then down
- ☐ Ghost going backward and then forward
- ☐ Ghost going up, flying forward for some steps and then coming down
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ghost going up, flying forward for some steps and then coming down

9) Which of the following is true?

1 point

- ☐ We can not implement an infinite loop using scratch
- ☐ We can not create a random integer using scratch
- ☐ Scratch does not have an explicit square root function
- ☐ Scratch does not have an explicit power function

No, the answer is incorrect.

Score: 0

Accepted Answers:

Scratch does not have an explicit power function

10) What does the following code compute

1 point



- ☐ multiplication of x and y
- ☐ x to the power y
- ☐ factorial of x
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

multiplication of x and y

X


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Unit 4 - Week 2

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

- ☒ Introduction to Anaconda (unit? unit=20&lesson=21)
- ☐ Installation of Anaconda (unit? unit=20&lesson=22)
- ☐ Introduction to Spyder IDE (unit? unit=20&lesson=23)
- ☒ Printing statements in Python (unit? unit=20&lesson=24)
- ☐ Understanding Variables in Python (unit? unit=20&lesson=25)
- ☐ Executing a sequence of

Assignment 2

The due date for submitting this assignment has passed. **Due on 2020-02-12, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Which of the following options correctly prints the phrase "Hurry Up!!"? 1 point

- ☐ printf("Hurry Up!!")
- ☐ print(Hurry Up!!!)
- ☐ print_sentence(Hurry Up!!!)
- ☐ print("Hurry Up!!")

No, the answer is incorrect.
Score: 0

Accepted Answers:
print("Hurry Up!!")

2) What is the output of the following code snippet? 1 point

```
1 a = "Hi "
2 b = "Arjuna "
3 c = "Bhimaa "
4 print("Hi", a, b, c)
```

- ☐ Hi Arjuna Bhimaa
- ☐ Hi Arjuna Hi Bhimaa
- ☐ Hi Hi Arjuna Bhimaa
- ☐ Hi Arjuna Bhimaa Hi

No, the answer is incorrect.
Score: 0

Accepted Answers:
Hi Hi Arjuna Bhimaa

instructions in the Console (unit? unit=20&lesson=26)

☐ Writing your First Program (unit? unit=20&lesson=27)

☐ Taking inputs from the user (unit? unit=20&lesson=28)

☐ Discount Calculation (unit? unit=20&lesson=29)

☐ Motivation to if condition (unit? unit=20&lesson=30)

☐ A reminder on how to deal with numbers (unit? unit=20&lesson=31)

☐ Understanding if condition's working (unit? unit=20&lesson=32)

☐ Realizing the importance of syntax and indentation (unit? unit=20&lesson=33)

☐ Introductions to loops (unit? unit=20&lesson=34)

☐ Loops: Sum of numbers (unit? unit=20&lesson=35)

☐ Loops: Sum of numbers (continued) (unit? unit=20&lesson=36)

☐ Loops: Multiplication Tables (unit? unit=20&lesson=37)

☐ Introduction to While Loop (unit? unit=20&lesson=38)

☐ Quiz : Assignment 2

3) With what extension are the python files saved?

1 point

- ☐ .p
☐ .python
☐ .pyn
☐ .py

No, the answer is incorrect.
 Score: 0

Accepted Answers:
 .py

4) Which of the following statements correctly represents taking input from user in python?

1 point

- ☐ a=get("Enter the value")
☐ a=inp("Enter the value")
☐ a=input("Enter the value")
☐ None of the above

No, the answer is incorrect.
 Score: 0

Accepted Answers:
 a=input("Enter the value")

5) What is the output of the following code?

1 point

```
1 d=-20
2 if (d>0)
3     print(d)
4 if (d<0):
5     print(-1*d)
```

- ☐ -20
☐ Syntax error
☐ 20
☐ Displays nothing

No, the answer is incorrect.
 Score: 0

Accepted Answers:
 Syntax error

6) What does the following code do?

1 point

```
1 for i in range(5):
2     print(i)
```

- ☐ displays numbers from 0 to 4
☐ displays numbers from 1 to 5
☐ displays numbers from 0 to 5
☐ none of the above

No, the answer is incorrect.
 Score: 0

Accepted Answers:
 displays numbers from 0 to 4

(assessment?
name=261)

- ☐ Programming Assignment-1: Introduction to Online Portal (/noc20_cs35/progassignment?name=268)
- ☐ Programming Assignment-2: Addition (/noc20_cs35/progassignment?name=269)
- ☐ Programming Assignment-3: Small (/noc20_cs35/progassignment?name=270)
- ☐ Week 2 Feedback (unit?unit=20&lesson=271)

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

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7) What does the following code do?

1 point

```
1 for i in range(10):
2     print("2 * ",(i) ,"=" ,2*(i+1))
```

- ☐ displays table of 2
- ☐ syntax error
- ☐ displays table of 10
- ☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
none of the above

8) What does the following code do?

1 point

```
1 a=1
2 for i in range(5):
3     a=a*(i+1)
4 print(a)
```

- ☐ 24
- ☐ 0
- ☐ 120
- ☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
120

9) What is the output of the following code?

1 point

```
1 d=5
2 while (d>1):
3     d=d-1
4     print(d) ,
```

- ☐ 4 3 2 1
- ☐ 5 4 3 2 1
- ☐ 4 3 2 1 0
- ☐ 5 4 3 2 1 0

No, the answer is incorrect.
Score: 0

Accepted Answers:
4 3 2 1

10)What is the output of the following code?

1 point

```
1 d="a"  
2 for i in range(2):  
3     d=d+1  
4     print(d) ,
```

- ☐ a b c
- ☐ a b
- ☐ b c
- ☐ Error

No, the answer is incorrect.

Score: 0

Accepted Answers:

Error

X


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Unit 5 - Week 3

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

- ☒ Lists Part 1 : Introduction (unit? unit=39&lesson=40)
- ☐ Lists Part 2 : Manipulation (unit? unit=39&lesson=41)
- ☐ Lists Part 3 : Operations (unit? unit=39&lesson=42)
- ☒ Lists Part 4 : Slicing (unit? unit=39&lesson=43)
- ☐ Loops and Conditionals : Fizzbuzz 01

Assignment 3

The due date for submitting this assignment has passed. **Due on 2020-02-19, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Which of the following displays a code which iterates from numbers 1 to 100, displays "fizz" if **1 point** the number is divisible by a but not b , displays "buzz" if the number is divisible by b but not a and displays "fizzbuzz" if the number is divisible by both a and b . a and b are inputs taken from the user.



```

1 a = int(input("Enter the value of a"))
2 b= int(input("Enter the value of b"))
3 for i in range(1,101):
4     if (i%a==0):
5         print("fizz")
6     if (i%b==0):
7         print("buzz")
8     else:
9         print("fizzbuzz")

```

(unit?
unit=39&lesson=44)

Loops and
Conditionals :
Fizzbuzz 02
(unit?
unit=39&lesson=45)

Crowd
Computing -
Just estimate 01
(unit?
unit=39&lesson=46)

Crowd
Computing -
Just estimate 02
(unit?
unit=39&lesson=47)

Crowd
Computing -
Just estimate 03
(unit?
unit=39&lesson=48)

Crowd
Computing -
Just estimate 04
(unit?
unit=39&lesson=49)

Crowd
Computing -
Just estimate 05
(unit?
unit=39&lesson=50)

Crowd
Computing -
Just estimate 06
(unit?
unit=39&lesson=51)

Permutations -
Jumbled Words
01 (unit?
unit=39&lesson=52)

Permutations -
Jumbled Words
02 (unit?
unit=39&lesson=53)

Permutations -
Jumbled Words
03 (unit?
unit=39&lesson=54)

Theory of
Evolution 01
(unit?
unit=39&lesson=55)

Theory of
Evolution 02

```

1 a = int(input("Enter the value of a"))
2 b= int(input("Enter the value of b"))
3 for i in range(1,101):
4     if (i%a==0):
5         if (i%b!=0):
6             print("fizz")
7         if (i%b==0):
8             if (i%a!=0):
9                 print("buzz")
10            else:
11                print("fizzbuzz")

```

```

1 a = int(input("Enter the value of a"))
2 b= int(input("Enter the value of b"))
3 for i in range(1,101):
4     if (i%a==0):
5         if (i%b!=0):
6             print("fizz")
7         if (i%b==0):
8             if (i%a!=0):
9                 print("buzz")
10            else:
11                print("fizzbuzz")

```

```

1 a = int(input("Enter the value of a"))
2 b= int(input("Enter the value of b"))
3 for i in range(1,101):
4     if (i%a==0):
5         if (i%b!=0):
6             print("fizz")
7         else:
8             if (i%a!=0):
9                 print("buzz")
10            else:
11                print("fizzbuzz")

```

No, the answer is incorrect.

Score: 0

Accepted Answers:

(unit?
unit=39&lesson=56)

☐ Theory of
Evolution 03
(unit?
unit=39&lesson=57)

☐ Theory of
Evolution 04
(unit?
unit=39&lesson=58)

☐ Quiz :
Assignment 3
(assessment?
name=262)

☐ Programming
Assignment-1:
Loops ,List and
Sum
(/noc20_cs35/progassignment?
name=273)

☐ Programming
Assignment-2:
Max and Min
(/noc20_cs35/progassignment?
name=274)

☐ Programming
Assignment-3:
Multiple of 5
(/noc20_cs35/progassignment?
name=275)

☐ Week 3
Feedback (unit?
unit=39&lesson=278)

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Text Transcripts

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```
1 a = int(input("Enter the value of a"))
2 b= int(input("Enter the value of b"))
3 for i in range(1,101):
4     if (i%a==0):
5         if (i%b!=0):
6             print("fizz")
7     if (i%b==0):
8         if (i%a!=0):
9             print("buzz")
10    else:
11        print("fizzbuzz")
```

2) The below code displays

1 point

```
1 for i in range(100):
2     if (i%10==0):
3         print(i)
```

- ☐ all numbers from 1 to 100 which are divisible by 10
- ☐ all numbers from 1 to 99 which are divisible by 10
- ☐ all numbers from 1 to 110 which are divisible by 10
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

none of the above

3) What is the output of the following code?

1 point

```
1 list1=["hi","we","are","the","elements","in","a","list"]
2 for i in list1:
3     print(list1[i]),
```

- ☐ hi hi hi hi hi hi hi
- ☐ hi we are the elements in a list
- ☐ hi
- ☐ error

No, the answer is incorrect.

Score: 0

Accepted Answers:

error

4) What is the output of the following code?

1 point

```
1 list1=["hi","we","are","the","elements","in","a","list"]
2 for i in range(4):
3     print(list1[i]),
```

Books

- ☐ hi we are the elements in a list
- ☐ hi we are the elements
- ☐ hi we are the
- ☐ hi we are

No, the answer is incorrect.

Score: 0

Accepted Answers:

hi we are the

5) Which of the following correctly represents a sum function in Python.

1 point

☐

```

1 def add(int a, int b):
2     print(a+b)
3 add(2,3)

```

☐

```

1 def add(a, b):
2     print(a+b)
3 add(2,3)

```

☐

```

1 def add(2, 3):
2     print(a+b)
3 add(a, b)

```

☐

```

1 function add(a, b):
2     print(a+b)
3 add(2,3)

```

No, the answer is incorrect.

Score: 0

Accepted Answers:

```

1 def add(a, b):
2     print(a+b)
3 add(2,3)

```

6) Which of the following statements correctly represents a jumbled form of a given word. For example, if the word="lion", one of the plausible outputs can be "oinl".

1 point

- ☐ "".join(random.sample(word,len(word)))
- ☐ "a".join(random.sample(word,len(word)))
- ☐ "".join(random.sample(word))
- ☐ join(random.sample(word))

No, the answer is incorrect.

Score: 0

Accepted Answers:

"".join(random.sample(word,len(word)))

7) Given a variable word which is a string where no letter is repeated, what does the following code snippet do?

1 point

```

1 import random
2 word=input("Enter a word in which no letter is repeated")
3 l=[]
4 total=1
5 for i in range(1,len(word)+1):
6     total=total*i
7 while (len(l)<total):
8     j="" .join(random.sample(word,len(word)))
9     l.append(j)
10 for each in l:
11     print each

```

- ☐ keeps printing the jumbled form of the word in an infinite loop
- ☐ shows all the jumbled forms of the word where some forms might be repeated
- ☐ shows some jumbled forms of the word where some forms might be repeated as well
- ☐ shows all unique jumbled forms of the word including the original word as well

No, the answer is incorrect.

Score: 0

Accepted Answers:

shows some jumbled forms of the word where some forms might be repeated as well

8) Assuming, there is no file named "file.txt" on my computer, what does the following code do? **1 point**

```

1 with open("file.txt","w") as f:
2     print(f.read())
3     f.write("Hey! I am writing");
4 f.close()

```

- ☐ Creates a file named file.txt and adds "Hey! I am writing" to it
- ☐ Shows an error because file does not exist
- ☐ Shows an error because file is not opened in the reading mode
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

None of the above

9) Assuming, there is no file named "file.txt" on my computer, what does the following code do? **1 point**

```

1 with open("file.txt","w") as f:
2     f.write("Hey! I am writing.");
3 f.close()
4 with open("file.txt","w") as f:
5     f.write("Hey I am writing the second line.");
6 f.close()
7 with open("file.txt","r") as f:
8     print(f.read())
9 f.close()

```

- ☐ Shows error
- ☐ Displays: Hey I am writing the second line
- ☐ Displays: Hey! I am writing.Hey I am writing the second line.
- ☐ Displays: Hey! I am writing.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Displays: Hey I am writing the second line

10) What is the output of the following code?

1 point

```
1 with open("f.txt","w") as myfile:
2     myfile.write("0110");
3 myfile.close()
4
5 with open("f.txt","r") as myfile:
6     l=list(myfile.read());
7     suml=0
8     for each in l:
9         suml=suml+int(each)
10    print(suml)
11 myfile.close()
```

- ☐ Displays 1
- ☐ Displays 2
- ☐ Displays 0
- ☐ Shows error

No, the answer is incorrect.

Score: 0

Accepted Answers:

Displays 2

X


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Unit 6 - week 4

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

- ☐ Practice is the key (unit? unit=59&lesson=60)
- ☐ Magic Square: Hit and Trial 01 (unit? unit=59&lesson=61)
- ☒ Magic Square: Hit and Trial 02 (unit? unit=59&lesson=62)
- ☐ Magic Square: Hit and Trial 03 (unit? unit=59&lesson=63)

Assignment 4

The due date for submitting this assignment has passed. **Due on 2020-02-26, 23:59 IST.**
As per our records you have not submitted this assignment.

1) What does the check_magic() function in the following code do

1 point

```

1  def check_magic():
2      num=[1,2,3,4,5,6,7,8,9]
3      a00=0
4      a01=0
5      a10=0
6      a11=0
7      for i in range(0,9):
8          for j in range(0,9):
9              for k in range(0,9):
10                 for l in range(0,9):
11                     a00=num[i]
12                     a01=num[j]
13                     a10=num[k]
14                     a11=num[l]
15                     l=[a00, a01, a10, a11]
16
17                 print a00, '\t', a01, '\n', a10, '\t', a11
18                 print '\n'

```

- ☐ displays all 2×2 matrices where elements are from 1 to 9.
- ☐ displays all 2×2 matrices where elements are from 1 to 9 but no element is repeated

☒ Magic Square:
Hit and Trial 04
(unit?
unit=59&lesson=64)

☒ Magic Square:
Hit and Trial 05
(unit?
unit=59&lesson=65)

☐ Let's program
and play (unit?
unit=59&lesson=66)

☐ Dobble Game -
Spot the
similarity 01
(unit?
unit=59&lesson=67)

☐ Dobble Game -
Spot the
similarity 02
(unit?
unit=59&lesson=68)

☒ Dobble Game -
Spot the
similarity 03
(unit?
unit=59&lesson=69)

☐ Dobble Game -
Spot the
similarity 04
(unit?
unit=59&lesson=70)

☒ What is your
date of birth?
(unit?
unit=59&lesson=71)

☐ Birthday
Paradox - Find
your twin 01
(unit?
unit=59&lesson=72)

☐ Birthday
Paradox - Find
your twin 02
(unit?
unit=59&lesson=73)

☐ Birthday
Paradox - Find
your twin 03
(unit?
unit=59&lesson=74)

☐ Birthday
Paradox - Find
your twin 04
(unit?
unit=59&lesson=75)

- ☐ displays magic squares of size 2
☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
displays all 2×2 matrices where elements are from 1 to 9.

2) What does the following code do?

1 point

```
1 l1 = ["apple", "banana", "kiwi", "orange"]
2 l2 = ["watermelon", "melon", "kiwi", "banana"]
3 cmn=[]
4 for i in range(4):
5     if (l1[i]==l2[i]):
6         cmn.append(l1[i])
7 print (cmn)
```

- ☐ displays common fruits in both the lists l1 and l2
☐ displays fruits which are in l1 but not in l2
☐ displays fruits which are in l2 but not in l1
☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
none of the above

3) Leap years are the years

1 point

1. which divisible by 4 but not divisible by 100, and, those
2. divisible by 400

Which of the following code does not represent a code displaying all the leap years from 1 to 2000.

☐

```
1 d4=[]
2 d100=[]
3 d400=[]
4 for i in range(1,2001):
5     if (i%4==0):
6         d4.append(i)
7     if (i%100==0):
8         d100.append(i)
9     if (i%400==0):
10        d400.append(i)
11 ly=[]
12 for each in d4:
13     if each not in d100:
14         ly.append(each)
15 for each in d400:
16     ly.append(each)
17 print (ly)
```

☐ Birthday
Paradox - Find
your twin 05
(unit?
unit=59&lesson=76)

☐ What's your
favourite movie?
(unit?
unit=59&lesson=77)

☐ Guess the
Movie Name 01
(unit?
unit=59&lesson=78)

☐ Guess the
Movie Name 02
(unit?
unit=59&lesson=79)

☐ Guess the
Movie Name 03
(unit?
unit=59&lesson=80)

☐ Guess the
Movie Name 04
(unit?
unit=59&lesson=81)

☐ Guess the
Movie Name 05
(unit?
unit=59&lesson=82)

☐ Guess the
Movie Name 06
(unit?
unit=59&lesson=83)

☐ **Quiz :**
Assignment 4
(assessment?
name=263)

☐ Programming
Assignment-1:
Digits
(/noc20_cs35/progassignment?
name=280)

☐ Programming
Assignment-2:
Factorial
(/noc20_cs35/progassignment?
name=281)

☐ Programming
Assignment-3:
Matrix
(/noc20_cs35/progassignment?
name=282)

☐ Week 4
Feedback (unit?
unit=59&lesson=283)

☐

```

1 ly=[]
2 for i in range(1,2001):
3     if (i%4==0):
4         if (i%100!=0):
5             ly.append(i)
6         else:
7             if (i%400==0):
8                 ly.append(i)
9 print(ly)

```

☐

```

1 ly=[]
2 for i in range(1,2001):
3     if (i%400==0):
4         ly.append(i)
5     else:
6         if (i%4==0):
7             ly.append(i)
8 print(ly)

```

☐

```

1 ly=[]
2 for i in range(1,2001):
3     if (i%400==0 or (i%100!=0 and i%4==0)):
4         ly.append(i)
5 print(ly)

```

No, the answer is incorrect.
Score: 0

Accepted Answers:

```

1 ly=[]
2 for i in range(1,2001):
3     if (i%400==0):
4         ly.append(i)
5     else:
6         if (i%4==0):
7             ly.append(i)
8 print(ly)

```

4) What does the following function do

1 point

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Text Transcripts

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Books

```

1 def leap(year):
2     if (year%400==0 or (year%100!=0 and year%4==0)):
3         return 1
4     else:
5         return 0

```

- ☐ returns true for century year and false for non century year
- ☐ returns true for leap year and false for non leap year
- ☐ returns false for century year and true for non century year
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

returns true for leap year and false for non leap year

5) Which of the following code correctly represents how one can display the number of dashes equal to that of the letters in the movie name? **1 point**

- ☐
- ```

movies=["titanic","chinatown","avengers","3idiots","conjuring","jungle
book","matrix"]
ch=random.choice(movies)
for i in range(len(ch)):
 print('_'),

```
- ☐
- ```

movies=["titanic","chinatown","avengers","3idiots","conjuring","jungle
book","matrix"]
ch=random.choice(movies)
for i in range(100):
    print('_'),

```
- ☐
- ```

movies=["titanic","chinatown","avengers","3idiots","conjuring","jungle
book","matrix"]
ch=random.choice(movies)
for ch in range(len(ch)):
 print('_'),

```
- ☐ none of these

No, the answer is incorrect.

Score: 0

Accepted Answers:

```

movies=["titanic","chinatown","avengers","3idiots","conjuring","junglebo
ok","matrix"]
ch=random.choice(movies)
for i in range(len(ch)):
 print('_'),

```

6) Given a list of movies, which of the following represents a code which randomly chooses a movie amongst all? **1 point**

- ☐
- ```

movies=["titanic","chinatown","avengers","3idiots","conjuring","jungle
book","matrix"]

```

- ☐ `ch = movies[random.randint(0, len(movies))]`
- ☐ `movies=["titanic","chinatown","avengers","3idiots","conjuring","junglebook","matrix"]`
`ch = movies[random.uniform(0, len(movies))]`
- ☐ `movies=["titanic","chinatown","avengers","3idiots","conjuring","junglebook","matrix"]`
`ch = movies[random.choice(0, len(movies))]`
- ☐ none of these

No, the answer is incorrect.

Score: 0

Accepted Answers:

none of these

7) What does the following code do?

1 point

```

1 s1=input("Enter a string")
2 s2=input("Enter another string")
3 for each in list(s2):
4     for each2 in list(s1):
5         if (each==each2):
6             print("yes")
7             break

```

- ☐ prints yes if both strings are same
- ☐ prints yes if both strings have atleast one common character
- ☐ prints yes if first string is contained in the second
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

prints yes if both strings have atleast one common character

8) Which numbers from 1 to 100 does the following code print?

1 point

```

1 for i in range(1001):
2     f=0
3     for j in range(2,i):
4         if (i%j==0):
5             f=1
6             break
7     if (f==0):
8         print(i)

```

- ☐ prime numbers
- ☐ perfect squares
- ☐ numbers which are factorial of some other number
- ☐ perfect cubes

No, the answer is incorrect.

Score: 0

Accepted Answers:

prime numbers

9) Which numbers from 1 to 100 does the following code print?

1 point

```

1  for i in range(1001):
2      f=0
3      for j in range(2,i):
4          if (j*j==i):
5              f=1
6              break
7      if (f==1):
8          print(i)

```

- ☐ prime numbers
☐ perfect squares
☐ numbers which are factorial of some other number
☐ perfect cubes

No, the answer is incorrect.

Score: 0

Accepted Answers:

perfect squares

10) Assume a drunkard whose movement is defined on the number line, i.e. he can either move forward or backward. Assume he is standing at a position p . He takes 2 steps forward followed by 4 steps backward. He falls into the pit as soon as he steps on the position zero. Which of the following codes correctly represents his walk? A. **1 point**

☐

```

p=int(input())
while (p > 0):
    p=p+2
    print(" Location =", p)
    p=p-4
    print(" Location =", p)
print(" Fell in pit at location ", p)

```

☐

```

p=int(input())
while (p > 0):
    p=p-2
    print(" Location =", p)
    p=p+4
    print(" Location =", p)
print(" Fell in pit at location ", p)

```

☐

```

p=int(input())
while (p > 0):
    for i in range(2):
        p=p+1
        print(" Loc =", p)

```

```

        if ( p ==0):
            break
    for i in range ( 4 ) :
        p=p-1
        print ( " Loc = " , p )
        if ( p ==0):
            break
    print (" Fell in pit at location " , p )

```

☐ none of these

No, the answer is incorrect.

Score: 0

Accepted Answers:

```

p= int ( input ( ) )
while ( p >0):
    for i in range ( 2 ) :
        p=p+1
        print ( " Loc = " , p )
        if ( p ==0):
            break
    for i in range ( 4 ) :
        p=p-1
        print ( " Loc = " , p )
        if ( p ==0):
            break
    print (" Fell in pit at location " , p )

```

X


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[swayam4@swayam.gov.in](#)
[NPTEL \(https://swayam.gov.in/explorer?ncCode=NPTEL\)](https://swayam.gov.in/explorer?ncCode=NPTEL) » [The Joy of Computing using Python \(course\)](#)
[Announcements \(announcements\)](#) [About the Course \(https://swayam.gov.in/nd1_noc20_cs35/preview\)](https://swayam.gov.in/nd1_noc20_cs35/preview)
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Unit 7 - Week 5

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

- ☒ Introduction to Dictionaries (unit? unit=84&lesson=85)
- ☐ Speech to Text : No need to write 01 (unit? unit=84&lesson=86)
- ☒ Speech to Text : No need to write 02 (unit? unit=84&lesson=87)
- ☒ Speech to Text : No need to write

Assignment 5

The due date for submitting this assignment has passed. **Due on 2020-03-04, 23:59 IST.**
As per our records you have not submitted this assignment.

1) What does the following code do?

1 point

```

1 import random
2 def get_gates():
3     r=random.randint(0,2)
4     r1=random.randint(0,2)
5     while(r==r1):
6         r=random.randint(0,2)
7     l=['x','x','x']
8     l[r]='c'
9     l[r1]='c'
10    ind=[0,1,2]
11    for each in ind:
12        if(each!=r1 and each!=r):
13            l[each]='g'
14    print(l)
15
16 get_gates()

```

- ☐ creates a list where two random elements are 'c' and the other element is 'g'
- ☐ creates a list where two random elements are 'g' and the other element is 'c'
- ☐ creates a list where one random elements is 'c' and the other element is 'g'
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

03 (unit?
unit=84&lesson=88)

☐ Monte Hall : 3
doors and a
twist 01 (unit?
unit=84&lesson=89)

☐ Monte Hall : 3
doors and a
twist 02 (unit?
unit=84&lesson=90)

☐ Rock, Paper
and Scissor :
Cheating not
allowed !! 01
(unit?
unit=84&lesson=91)

☐ Rock, Paper
and Scissor :
Cheating not
allowed !! 02
(unit?
unit=84&lesson=92)

☐ Rock, Paper
and Scissor :
Cheating not
allowed !! 03
(unit?
unit=84&lesson=93)

☐ Rock, Paper
and Scissor :
Cheating not
allowed !! 04
(unit?
unit=84&lesson=94)

☐ Sorting and
Searching : 20
questions game
01 (unit?
unit=84&lesson=95)

☐ Sorting and
Searching : 20
questions game
02 (unit?
unit=84&lesson=96)

☐ Sorting and
Searching : 20
questions game
03 (unit?
unit=84&lesson=97)

☐ Sorting and
Searching : 20
questions game
04 (unit?
unit=84&lesson=98)

☐ Sorting and
Searching : 20

creates a list where two random elements are 'c' and the other element is 'g'

2) Which of the random experiments from the options does the code represent?

1 point

```
1 import random
2 while (1):
3     r=random.randint(0,1)
4     if (r==0):
5         print('tossing')
6         break
7     else:
8         print('tossing')
```

- ☐ Tossing a coin once
- ☐ Tossing a coin infinite times
- ☐ Tossing a coin repeatedly till a head is encountered
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

none of the above

3) Which of the random experiments from the options does the code represent?

1 point

```
1 import random
2 p1=["rock","paper","scissor"]
3 p2=["rock","paper","scissor"]
4 c1=random.choice(p1)
5 c2=random.choice(p2)
6 if (c1==c2):
7     print("SUCCESS")
8 else:
9     print("FAIL")
```

- ☐ Prints a success when both people select the same object
- ☐ Prints a success when both people select "rock"
- ☐ Prints a success when both people select different objects
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Prints a success when both people select the same object

4) For the code below, which of the statement in the options is false?

1 point

```
1 t=[]
2 for i in range(10):
3     a=int(input("Enter the number you want to insert in the list"))
4     if (len(t)==0):
5         t.append(a)
6     else:
7         if (a>t[len(t)-1]):
8             t.append(a)
9     print(t)
```

questions game
05 (unit?
unit=84&lesson=99)

☐ Sorting and
Searching : 20
questions game
06 (unit?
unit=84&lesson=100)

☐ Sorting and
Searching : 20
questions game
07 (unit?
unit=84&lesson=101)

☐ Sorting and
Searching : 20
questions game
08 (unit?
unit=84&lesson=102)

☐ Quiz :
Assignment 5
(assessment?
name=264)

☐ Programming
Assignment-1:
Cab and walk
(/noc20_cs35/progassignment?
name=291)

☐ Programming
Assignment-2:
End-Sort
(/noc20_cs35/progassignment?
name=292)

☐ Programming
Assignment-3:
Semi Primes
(/noc20_cs35/progassignment?
name=293)

☐ Week 5
Feedback (unit?
unit=84&lesson=294)

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

- ☐ The loop runs exactly 10 times
- ☐ All the integers taken as input from the user need not be in the list l
- ☐ The list l consists of exactly 10 elements at the end of the program
- ☐ The list l printed in the last line is a sorted list

No, the answer is incorrect.
Score: 0

Accepted Answers:

The list l consists of exactly 10 elements at the end of the program

5) Which of the random experiments from the options does the code represent? **1 point**

```
1 import random
2 bins={}
3 for i in range(1,11):
4     bins[i]=0
5 for i in range(1,101):
6     r = random.randint(1,10)
7     bins[r]=bins[r]+1
8 print(bins)
```

- ☐ Placing 100 bins and then throwing 10 balls randomly in these bins
- ☐ Placing 10 bins and then throwing 100 balls randomly in these bins
- ☐ Placing 10 bins and 10 balls and then throwing 10 balls randomly in these bins
- ☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

Placing 10 bins and then throwing 100 balls randomly in these bins

6) Assuming that "bins" represents a dictionary where key is the number of a bin and value represents the number of balls present in the corresponding bin, what is the output of the following code? **1 point**

```
1 min_=0
2 min_i=-1
3 for each in bins:
4     if (bins[each]>min_):
5         min_i=each
6         min_=bins[each]
7 print(min_i)
```

- ☐ Displays the maximum number of balls present in any bin
- ☐ Displays the number of the bin containing maximum balls
- ☐ Displays the number of the bin containing minimum balls
- ☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

Displays the number of the bin containing maximum balls

7) Assuming that "bins" represents a dictionary where key is the number of a bin and value represents the number of balls present in the corresponding bin, what is the output of the following code? **1 point**

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```

1 def mbin():
2     max_=0
3     max_i=-1
4     for each in bins:
5         if (bins[each]>max_):
6             max_i=each
7             max_=bins[each]
8     print(max_i)
9     return max_i
10
11 while(len(bins)>0):
12     b=mbin()
13     del(bins[b])

```

- ☐ Displays the maximum number of balls present in any bin
- ☐ Displays bins in the ascending order of the number of balls they have
- ☐ Displays bins in the descending order of the number of balls they have
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Displays bins in the descending order of the number of balls they have

8)

1 point

```

1 def find(list1,num):
2     for each in list1:
3         if(each!=num):
4             print(each)
5         else:
6             break
7
8 t=[]
9 for i in range(100000):
10     t.append(i)
11
12 find(t,99999)

```

The above code generates numbers from

- ☐ 0 to 99999
- ☐ 0 to 100000
- ☐ 0 to 99998
- ☐ 1 to 99998

No, the answer is incorrect.

Score: 0

Accepted Answers:

0 to 99998

9) Which of the random experiments from the options does the code represent?

1 point

```

1 import random
2 while (1):
3     r=random.randint(1,6)
4     if (r%2==0):
5         print('rolling')
6         break
7     else:
8         print('rolling')

```

- ☐ Rolling a dice once
- ☐ Rolling a dice infinite times
- ☐ Rolling a dice repeatedly till an odd number is encountered
- ☐ Rolling a dice repeatedly till an even number is encountered

No, the answer is incorrect.

Score: 0

Accepted Answers:

Rolling a dice repeatedly till an even number is encountered

10) Assuming that “bins” represents a dictionary where key is the number of a bin and value represents the number of balls present in the corresponding bin, what plot does the following code generate?

1 point

```

1 import matplotlib.pyplot as plt
2 val=bins.values()
3 x=[]
4 y=[]
5 print(val)
6 for each in list(set(val)):
7     x.append(each)
8     y.append(val.count(each))
9     print(each, val.count(each))
10 plt.plot(x,y)
11 plt.show()

```

- ☐ X axis: Number of balls, Y axis: Number of bins having as many balls as specified by X axis
- ☐ X axis: Bin number, Y axis: Number of balls in the bin whose number is specified by X axis
- ☐ X axis: Ball number, Y axis: The bin number which contained the ball whose number is specified by the X axis
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

None of the above

X


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https://swayam.gov.in/nc_details/NPTEL

[REDACTED]

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » **The Joy of Computing using Python (course)**

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Unit 8 - Week 6

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

☒ Substitution Cipher -The science of secrecy (unit? unit=103&lesson=104)

☐ Substitution Cipher -The science of secrecy 01 (unit? unit=103&lesson=105)

☐ Substitution Cipher -The

Assignment 6

The due date for submitting this assignment has passed. **Due on 2020-03-11, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Give that the statement `chr(ord(alpha) + i)` returns the character(alphabet or a special character) at the location `i` ahead than the alphabet `alpha`, eg, `chr(ord('a')+1)` returns 'b'; what is the output of the following code?

1 point

```
1 def encrypt(ltr, key):
2     l=[]
3     for each in list(ltr):
4         l.append(chr(ord(each) + 1))
5     return "".join(l)
6
7 letter_body="ABCDEFGH"
8 d=encrypt(letter_body, 4)
9 print(d)
```

- ☐ ABCDEFGH
- ☐ BCDEFGHI
- ☐ EFGHIJKL
- ☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
BCDEFGHI

2) What does the following code do?

1 point

science of
secrecy 02
(unit?
unit=103&lesson=106)

☐ Substitution
Cipher -The
science of
secrecy 03
(unit?
unit=103&lesson=107)

☐ Tic Tac Toe -
Down the
memory Lane
(unit?
unit=103&lesson=108)

☐ Tic Tac Toe -
Down the
memory Lane
01 (unit?
unit=103&lesson=109)

☐ Tic Tac Toe -
Down the
memory Lane
02 (unit?
unit=103&lesson=110)

☐ Tic Tac Toe -
Down the
memory Lane
03 (unit?
unit=103&lesson=111)

☐ Tic Tac Toe -
Down the
memory Lane
04 (unit?
unit=103&lesson=112)

☐ Tic Tac Toe -
Down the
memory Lane
05 (unit?
unit=103&lesson=113)

☐ Recursion (unit?
unit=103&lesson=114)

☐ Recursion 01
(unit?
unit=103&lesson=115)

☐ Recursion 02
(unit?
unit=103&lesson=116)

☐ Recursion 03
(unit?
unit=103&lesson=117)

☐ Recursion 04
(unit?
unit=103&lesson=118)

```
1 def guess(num):
2     a=input("Guess a number")
3     if (a==num):
4         print("SUCCESS")
5     else:
6         guess(num)
7
8 guess(10)
```

- ☐ Keeps asking the user to guess a number until the user guesses 10
- ☐ The computer generates a random number r and keeps it. The user is repeatedly prompted to enter a number. If the user enters r, the code says success and ends, else the prompting is continued.
- ☐ Enters an infinite loop
- ☐ The computer generates a random number r and keeps it. The user is repeatedly prompted to enter a number. If the user enters r, the code says success and ends, else the computer generates a new random number r and thereafter the prompting is continued.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Enters an infinite loop

3) What does the following code do?

1 point

```
1 import random
2 def guess(num):
3     a=int(input("Guess a number from 1 to 100"))
4     print(a,num)
5     if (a==num):
6         print("SUCCESS")
7     else:
8         guess(random.randint(1,100))
9
10 i=guess(random.randint(1,100))
```

- ☐ Keeps asking the user to guess a number until the user guesses 10
- ☐ The computer generates a random number r and keeps it. The user is repeatedly prompted to enter a number. If the user enters r, the code says success and ends, else the prompting is continued.
- ☐ Enters an infinite loop
- ☐ The computer generates a random number r and keeps it. The user is repeatedly prompted to enter a number. If the user enters r, the code says success and ends, else the computer generates a new random number r and thereafter the prompting is continued.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The computer generates a random number r and keeps it. The user is repeatedly prompted to enter a number. If the user enters r, the code says success and ends, else the computer generates a new random number r and thereafter the prompting is continued.

4) With n as input, the code below computes

1 point

☐ Recursion 05
(unit?
unit=103&lesson=119)

☐ Recursion 06
(unit?
unit=103&lesson=120)

☐ Quiz :
Assignment 6
(assessment?
name=276)

☐ Programming
Assignment-1:
Computing
Paradox
(/noc20_cs35/progassignment?
name=295)

☐ Programming
Assignment-2:
Dictionary
(/noc20_cs35/progassignment?
name=296)

☐ Programming
Assignment-3:
Functions
(/noc20_cs35/progassignment?
name=297)

☐ Week 6
Feedback (unit?
unit=103&lesson=298)

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

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```
1 def mul(num):
2     if (num==1):
3         return (-1)
4     return (-1*mul(num-1))
5
6 n=int(input("Enter the value of n"))
7 print(mul(n))
```

☐ $-1 \times n$

☐ $-1 + n$

☐ $(-1)^n$

☐ $n^{(-1)}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-1)^n$

5) The following code

1 point

```
1 import random
2 def search(l, loc, item):
3     if (loc < 0):
4         loc = 0
5     if (l[loc] == item):
6         print("Found", item, "at index", loc)
7         return
8     if (loc == len(l) - 1):
9         print("Element not present")
10        return (0)
11    else:
12        return (search(l, loc+1, item))
13
14 l = [1, 2, 3, 4, 5, 6, 7, 8, 9]
15 search(l, -11, 3)
```

☐ displays an error

☐ does not display an error but might display the error if we change the middle value passed in the function search() from 0 to some negative value.

☐ Can return a negative value in some cases when we change the values passed to the function search()

☐ Scans the list from first to the last element and displays the index of the value passed in the last number in the function search().

No, the answer is incorrect.

Score: 0

Accepted Answers:

Scans the list from first to the last element and displays the index of the value passed in the last number in the function search().

6) The following code represents

1 point

```

1 import random
2 def search(l, loc, item):
3     if (loc < 0):
4         loc = 0
5     if (l[loc] == item):
6         print("Found", item, "at index", loc)
7         return
8     if (loc == len(l) - 1):
9         print("Element not present")
10        return (0)
11    else:
12        return (search(l, loc + 1, item))
13
14 l = [1, 2, 3, 4, 5, 6, 7, 8, 9]
15 search(l, -11, 3)

```

- ☐ recursive algorithm for linear search an element in a list
☐ recursive algorithm for binary search an element in a list
☐ non-recursive algorithm for linear search an element in a list
☐ none of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

recursive algorithm for linear search an element in a list

7) What is the output of `print(int(3.79)+int(2.1))`?

1 point

- ☐ 6
☐ 5
☐ 7
☐ 4

No, the answer is incorrect.
Score: 0

Accepted Answers:

5

8) The following code to its best, represents a scenario

1 point

```

1 def func(i):
2     print(i)
3     if (i == 0):
4         print("OVER")
5     else:
6         func(i / 2)

```

- ☐ A cake getting eaten by half of its current amount every time
☐ A student attempting alternate questions, starting from a given question
☐ Viruses doubling inside a body and killing the person once their population becomes 128 or more.
☐ Metro train serving 128 stations to and fro

No, the answer is incorrect.
Score: 0

Accepted Answers:

A cake getting eaten by half of its current amount every time

9) The following code to its best, represents a scenario

1 point

```

1 def func(i):
2     print(i)
3     if (i>128):
4         print("OVER")
5     else:
6         func(2*i)

```

- ☐ A cake getting eaten by half of its current amount every time
- ☐ A student attempting alternate questions, starting from a given question
- ☐ Viruses doubling inside a body and killing the person once their population becomes 128 or more.
- ☐ Metro train serving 128 stations to and fro

No, the answer is incorrect.

Score: 0

Accepted Answers:

Viruses doubling inside a body and killing the person once their population becomes 128 or more.

10) The following code to its best, represents a scenario

1 point

```

1 def func(i, f):
2     print(i)
3     if (i==0):
4         f=1
5         func(i+1, f)
6     if (i==128):
7         f=-1
8         func(i-1, f)
9     if (f==1):
10        func(i+1, f)
11    if (f==-1):
12        func(i-1, f)

```

- ☐ A cake getting eaten by half of its current amount every time
- ☐ A student attempting alternate questions, starting from a given question
- ☐ Viruses doubling inside a body and killing the person once their population becomes 128 or more.
- ☐ Metro train serving 128 stations to and fro

No, the answer is incorrect.

Score: 0

Accepted Answers:

Metro train serving 128 stations to and fro

X


<https://swayam.gov.in>

https://swayam.gov.in/nc_details/NPTEL

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » **The Joy of Computing using Python (course)**

Announcements (announcements) **About the Course** (https://swayam.gov.in/nd1_noc20_cs35/preview)

Ask a Question (forum) Progress (student/home) Mentor (student/mentor)

Unit 9 - Week 7

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

☒ Snakes and Ladders - Not on the Board (unit? unit=121&lesson=122)

☐ Snakes and Ladders - Not on the Board - Part 01 (unit? unit=121&lesson=123)

Assignment 7

The due date for submitting this assignment has passed. **Due on 2020-03-18, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Predict the output

1 point

```

1 l=[[1,2,3],[4,5,6],[7,8,9]]
2 f=1
3 for j in range(3):
4     if (f==1):
5         for i in range(3):
6             print(l[i][j],end=" ")
7             f=0
8     if (f==0):
9         for i in range(2,-1,-1):
10            print(l[i][j],end=" ")
11            f=1

```

☐ 7 4 1 1 4 7 2 5 8 8 5 2 3 6 9 9 6 3

☐ 1 4 7 2 5 8 3 6 9

☐ 1 4 7 8 5 2 3 6 9

☐ 1 4 7 7 4 1 2 5 8 8 5 2 3 6 9 9 6 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

1 4 7 7 4 1 2 5 8 8 5 2 3 6 9 9 6 3

2) Predict the output of the calling function func1() for a given square matrix mx of dimension 70 × 70. **1 point**

☐ Snakes and Ladders - Not on the Board - Part 02 (unit? unit=121&lesson=124)

☐ Snakes and Ladders - Not on the Board - Part 03 (unit? unit=121&lesson=125)

☐ Snakes and Ladders - Not on the Board - Part 04 (unit? unit=121&lesson=126)

☐ Snakes and Ladders - Not on the Board - Part 05 (unit? unit=121&lesson=127)

☐ Snakes and Ladders - Not on the Board - Part 06 (unit? unit=121&lesson=128)

☐ Spiral Traversing - Let's Animate (unit? unit=121&lesson=129)

☐ Spiral Traversing - Let's Animate - Part 01 (unit? unit=121&lesson=130)

☐ Spiral Traversing - Let's Animate - Part 02 (unit? unit=121&lesson=131)

☐ Spiral Traversing - Let's Animate - Part 03 (unit? unit=121&lesson=132)

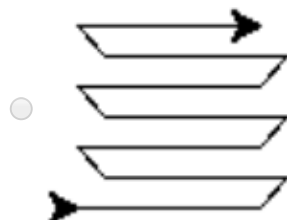
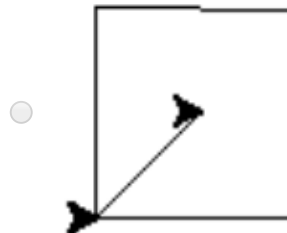
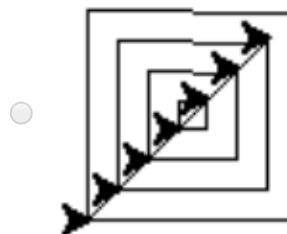
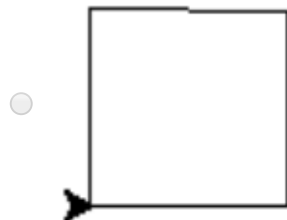
☐ Spiral Traversing - Let's Animate - Part 04 (unit? unit=121&lesson=133)

☐ Spiral Traversing - Let's Animate - Part 05 (unit? unit=121&lesson=134)

```

1  def func(mx,i):
2      tur = turtle.Turtle()
3      tur.setpos(i,i)
4
5      for ind in range(i,n-i):
6          tur.goto(i,ind)
7      for ind in range(i+1,n-i):
8          tur.goto(i,n-1-i)
9      for ind in range(n-2-i,i,-1):
10         tur.goto(n-1-i,ind)
11     for ind in range(n-i-1,i,-1):
12         tur.goto(ind,i)
13
14     def func1(mx):
15         n=len(mx)
16         i=0
17         while (i <=n-1):
18             func(mx,i)
19             i=i+10
20

```



No, the answer is incorrect.
Score: 0
Accepted Answers:

☐ Spiral Traversing - Let's Animate - Part 06 (unit? unit=121&lesson=135)

☐ Spiral Traversing - Let's Animate - Part 07 (unit? unit=121&lesson=136)

☐ GPS - Track the route (unit? unit=121&lesson=137)

☐ GPS - Track the route - Part 01 (unit? unit=121&lesson=138)

☐ GPS - Track the route - Part 02 (unit? unit=121&lesson=139)

☐ GPS - Track the route - Part 03 (unit? unit=121&lesson=140)

☐ GPS - Track the route - Part 04 (unit? unit=121&lesson=141)

☐ **Quiz : Assignment 7 (assessment? name=277)**

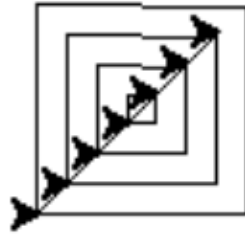
☐ Programming Assignment-1: Lower Triangular Matrix (/noc20_cs35/progassignment? name=299)

☐ Programming Assignment-2: Symmetric (/noc20_cs35/progassignment? name=300)

☐ Programming Assignment-3: Binary Matrix (/noc20_cs35/progassignment? name=301)

☐ Week 7 Feedback (unit? unit=121&lesson=302)

Week 8

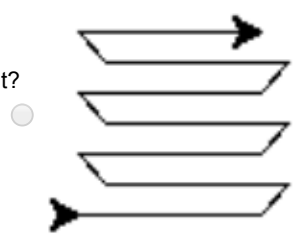
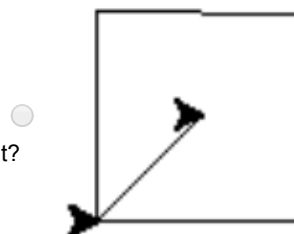
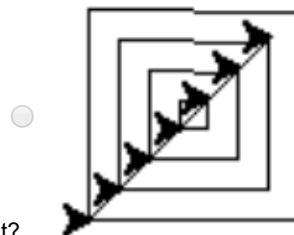
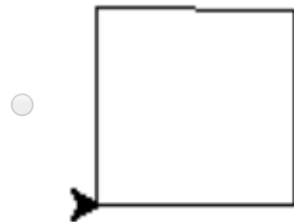


3) Predict the output of the calling function func() for a given square matrix mx of dimension 70 × 70. **0 points**

```

1 def func(mx):
2     func1(mx,0)
3     tur = turtle.Turtle()
4     tur.setpos(0,0)
5     if ((len(mx))%2==1):
6         turtle.goto(int(len(mx)/2),int(len(mx)/2))
7     else:
8         second=int(len(mx)/2)
9         turtle.goto(second-1,second-1)
10        turtle.goto(second-1,second)
11        turtle.goto(second,second-1)
12        turtle.goto(second,second)
13

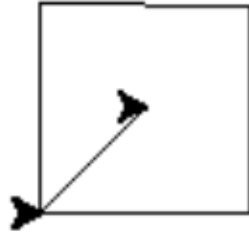
```



No, the answer is incorrect.
Score: 0

Week 9**Week 10****Week 11****Week 12****Text Transcripts****Download Videos****Books**

Accepted Answers:



4) Which of the following libraries is required to work with Google maps in Python?

1 point

- ☐ gplot
☐ googleplot
☐ gmplot
☐ none of these

No, the answer is incorrect.

Score: 0

Accepted Answers:

gmplot

5) Which of the following codes represent a correct version of a board game where the user has **1 point** to move from block 1 to block 100?

The game initialises only when the user gets a 1 or 6 on the dice and ends once he reaches 100 or gets a number which makes

him reach beyond 100 (i.e. the player wins if he is at 99 and gets a 4).



```

1 import random
2 def play(psn):
3     r = random.randint(1,6)
4     if (psn==0):
5         if (r==1 or r==6):
6             psn=1
7         else:
8             psn=psn+r
9     print("Position=",psn)
10    if (psn>=100):
11        print("You won")
12        return
13    play(psn)
14    position=0
15    print("Position=",position)
16    play(position)
  
```



```

1 import random
2
3 def play(psn):
4
5     r = random.randint(1,6)
6     print("Dice rolled:",r)
7     if(psn==0):
8         if(r==1 or r==6):
9             psn=1
10        else:
11            psn=psn+r
12        print("Position=",psn)
13        if(psn>=100):
14            print("You won")
15            return
16        play(psn)
17
18
19 position=0
20 print("Position=",position)
21 play(position)

```



```

1 import random
2
3 def play(psn):
4
5     r = random.randint(1,6)
6     print("Dice rolled:",r)
7     input()
8     if(psn==0):
9         if(r==1 or r==6):
10            psn=1
11        else:
12            psn=psn+r
13        print("Position=",psn)
14        if(psn>=100):
15            print("You won")
16
17        play(psn)
18
19
20 position=0
21 print("Position=",position)
22 play(position)

```



```

1 import random
2
3 def play(psn):
4

```

```

5     print("Dice rolled:",2)
6     if (psn==0):
7         psn=1
8     else:
9         psn=psn+2
10    print("Position=",psn)
11    if (psn>=100):
12        print("You won")
13
14    play(psn)
15 position=0
16 print("Position=",position)
17 play(position)

```

No, the answer is incorrect.

Score: 0

Accepted Answers:

```

1 import random
2
3 def play(psn):
4
5     r = random.randint(1,6)
6     print("Dice rolled:",r)
7     if (psn==0):
8         if (r==1 or r==6):
9             psn=1
10    else:
11        psn=psn+r
12    print("Position=",psn)
13    if (psn>=100):
14        print("You won")
15        return
16    play(psn)
17
18
19 position=0
20 print("Position=",position)
21 play(position)

```

6) Imagine a single player snakes and ladders game. The code below represents

1 point

```

1 import random
2
3 def play(psn):
4     snake_begin=-1
5     snake_end=-1
6     while(snake_begin <= snake_end):
7         snake_begin=random.randint(1,99)
8         snake_end=random.randint(1,99)
9         print("Snake from",snake_begin,"to",snake_end)
10        r = random.randint(1,6)
11        print("Dice rolled:",r)
12        if(psn==0):
13            if(r==1 or r==6):
14                psn=1
15        else:
16            psn=psn+r
17        print("Position=",psn)
18        input()
19        if(psn==snake_begin):
20            print("Bitten by snake")
21            psn=snake_end
22        if(psn>=100):
23            print("You won")
24            return
25        play(psn)
26
27 position=0
28 print("Position=",position)
29 play(position)

```

- ☐ A snakes and ladders game with one snake whose position remains constant while the player is playing. The position also remains the same during any subsequent plays (i.e. the game board does not change while you sleep and play again the next day).
- ☐ A snakes and ladders game with one snake whose position remains constant while the player is playing. However, the position can change during any subsequent plays (i.e. the game board might change while you sleep and play again the next day).
- ☐ A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snakes keep moving). Further, the snake can bite you any number of times.
- ☐ A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snake keeps moving). Further, the snake can bite you only ones when you play.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snakes keep moving). Further, the snake can bite you any number of times.

7) Imagine a single player snakes and ladders game. The code below represents

1 point

```

1 import random
2
3 def play(psn, flag):
4     snake_begin=-1
5     snake_end=-1
6     while(snake_begin <= snake_end):
7         snake_begin=random.randint(1,99)
8         snake_end=random.randint(1,99)
9         print("Snake from", snake_begin, "to", snake_end)
10        r = random.randint(1,6)
11        print("Dice rolled:", r)
12        if(psn==0):
13            if(r==1 or r==6):
14                psn=1
15        else:
16            psn=psn+r
17        print("Position=", psn)
18        #input()
19        if(psn==snake_begin and flag==0):
20            print("Bitten by snake")
21            psn=snake_end
22            flag=1
23        if(psn>=100):
24            print("You won")
25            return
26        play(psn, flag)
27
28 position=0
29 print("Position=", position)
30 play(position, 0)

```

- ☐ A snakes and ladders game with one snake whose position remains constant while the player is playing. The position also remains the same during any subsequent plays (i.e. the game board does not change while you sleep and play again the next day).
- ☐ A snakes and ladders game with one snake whose position remains constant while the player is playing. However, the position can change during any subsequent plays (i.e. the game board might change while you sleep and play again the next day).
- ☐ A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snakes keep moving). Further, the snake can bite you any number of times.
- ☐ A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snake keeps moving). Further, the snake can bite you only ones when you play.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snake keeps moving). Further, the snake can bite you only ones when you play.

8) Assuming that the play1() function implements the recursive play of snakes and ladders with **1 point** the prespecified position of the snake and the ladder as shown in the code below, which of the ambiguities in the options can result in the code?

Kindly assume a typical snakes and ladders game.

```

1 import random
2 def play(psn):
3     snake_begin=-1
4     snake_end=-1
5     while(snake_begin <= snake_end):
6         snake_begin=random.randint(1,99)
7         snake_end=random.randint(1,99)
8     ladder_begin=-1
9     ladder_end=-1
10    while(ladder_end <= ladder_begin):
11        ladder_begin=random.randint(1,99)
12        ladder_end=random.randint(1,99)
13    play1(psn, snake_begin, snake_end, ladder_begin, ladder_end)

```

- ☐ snake_begin=snake_end
☐ ladder_begin=ladder_end
☐ ladder_begin=snake_begin
☐ ladder_end=snake_end

No, the answer is incorrect.

Score: 0

Accepted Answers:

ladder_begin=snake_begin

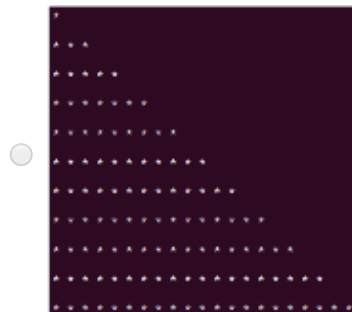
9) What is the output of the following code?

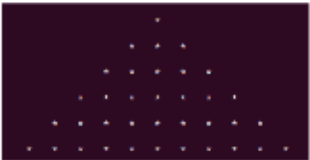
0 points

```

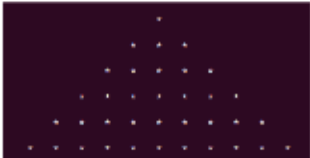
1 import random
2 def play(psn):
3     snake_begin=-1
4     snake_end=-1
5     while(snake_begin <= snake_end):
6         snake_begin=random.randint(1,99)
7         snake_end=random.randint(1,99)
8     ladder_begin=-1
9     ladder_end=-1
10    while(ladder_end <= ladder_begin):
11        ladder_begin=random.randint(1,99)
12        ladder_end=random.randint(1,99)
13    play1(psn, snake_begin, snake_end, ladder_begin, ladder_end)

```



- ☐ 
- ☐ 
- ☐ 

No, the answer is incorrect.
Score: 0
Accepted Answers:



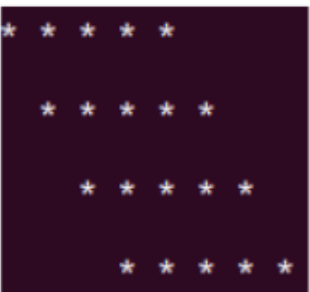
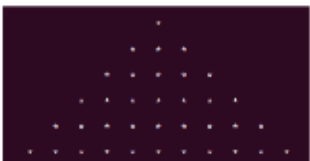
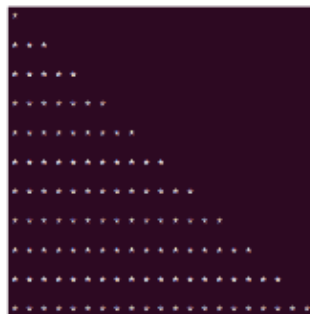
10)What is the output of the following code?

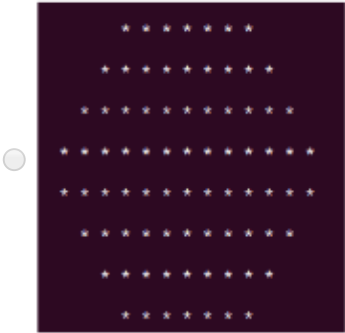
1 point

```

1
2 def func():
3     print()
4     c=10
5     i=3
6     while(i <=6):
7         j=0
8         while(j <=20):
9             if(j >=10-i and j <=10+i):
10                print('*',end=" ")
11            else:
12                print(' ',end=" ")
13            j=j+1
14        print('\n')
15        i=i+1
16    i=6
17    while(i >=3):
18        j=0
19        while(j <=20):
20            if(j >=10-i and j <=10+i):
21                print('*',end=" ")
22            else:
23                print(' ',end=" ")
24            j=j+1
25        print('\n')
26        i=i-1
27 func()

```





No, the answer is incorrect.
Score: 0
Accepted Answers:



X


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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » **The Joy of Computing using Python (course)**

Announcements (announcements) **About the Course** (https://swayam.gov.in/nd1_noc20_cs35/preview)

Ask a Question (forum) Progress (student/home) Mentor (student/mentor)

Unit 10 - Week 8

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

☐ Tuples- Python Data Structure (unit? unit=142&lesson=143)

☐ Lottery Simulation - Profit or Loss (unit? unit=142&lesson=144)

Assignment 8

The due date for submitting this assignment has passed. **Due on 2020-03-25, 23:59 IST.**
As per our records you have not submitted this assignment.

Note that Q8 carries 2 marks.

1) Which of the following options correctly represent the full form of acronyms NLTK and VADER **1 point**

- ☐ NLTK: Normal Language Toolkit, VADER: Valence Aware Dictionary and Emotional Reasoner
- ☐ NLTK: Natural Language Toolkit, VADER: Valence Aware Dictionary and Sentiment Reasoner
- ☐ NLTK: Normal Language Toolkit, VADER: Valence Aware Dictionary and Sentiment Reasoner
- ☐ Natural Language Toolkit, VADER: Valence Aware Dictionary and Emotional Reasoner

No, the answer is incorrect.

Score: 0

Accepted Answers:

NLTK: Natural Language Toolkit, VADER: Valence Aware Dictionary and Sentiment Reasoner

2) Predict the output

1 point

```
1 string1="HI! Amitabh"
2 print(sorted(string1))
```

- ☐ [' ', 'I', 'A', 'H', 'I', 'a', 'b', 'h', 'I', 'm', 't']
- ☐ ['I', 'A', 'H', 'I', 'a', 'b', 'h', 'I', 'm', 't']
- ☐ !AHIabhimt
- ☐ !AabHhlimt

No, the answer is incorrect.

Score: 0

Accepted Answers:

[' ', 'I', 'A', 'H', 'I', 'a', 'b', 'h', 'I', 'm', 't']

- ☐ Lottery
Simulation -
Profit or Loss -
Part 01 (unit?
unit=142&lesson=145)
- ☐ Lottery
Simulation -
Profit or Loss -
Part 02 (unit?
unit=142&lesson=146)
- ☐ Lottery
Simulation -
Profit or Loss -
Part 03 (unit?
unit=142&lesson=147)
- ☐ Lottery
Simulation -
Profit or Loss -
Part 04 (unit?
unit=142&lesson=148)
- ☐ Lottery
Simulation -
Profit or Loss -
Part 05 (unit?
unit=142&lesson=149)
- ☐ Lottery
Simulation -
Profit or Loss -
Part 06 (unit?
unit=142&lesson=150)
- ☐ Image
Processing -
Enhance your
images (unit?
unit=142&lesson=151)
- ☐ Image
Processing -
Enhance your
images - Part 01
(unit?
unit=142&lesson=152)
- ☐ Image
Processing -
Enhance your
images - Part 02
(unit?
unit=142&lesson=153)
- ☐ Image
Processing -
Enhance your
images - Part 03
(unit?
unit=142&lesson=154)
- ☐ Anagrams (unit?
unit=142&lesson=155)

3) Which of the scenarios in the options does the following code represent?

1 point

```

1 import random
2 def play():
3     a=input("Enter a number from 1 to 10")
4     r=random.randint(1,10)
5     if(a==r):
6         return 1
7     else:
8         return 0
9
10 amt=0
11 for i in range(1,366):
12     amt=amt+play()
13
14 print(amt)

```

- ☐ A person going to the bar for an year. Daily he guesses a number from 1 to 10. If the guessed number if equal to the number randomly generated by bar authority, he gains one gold coin.
- ☐ A person going to the bar for a month. Daily he guesses a number from 1 to 10. If the guessed number if equal to the number randomly generated by bar authority, he gains one gold coin.
- ☐ A person going to the bar for an year. Daily he guesses a number from 1 to 10. If the guessed number if equal to the number randomly generated by bar authority, he loses one gold coin.
- ☐ A person going to the bar for a month. Daily he guesses a number from 1 to 10. If the guessed number if equal to the number randomly generated by bar authority, he loses one gold coin.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A person going to the bar for an year. Daily he guesses a number from 1 to 10. If the guessed number if equal to the number randomly generated by bar authority, he gains one gold coin.

4) Which of the scenarios in the options does the following code represent?

1 point

```

1 import random
2
3
4 def play():
5     amt=0
6     for i in range(0,100):
7         r=random.uniform(0,1)
8         if(r<0.5):
9             amt=amt+1
10    return amt
11
12
13 s=0
14 for i in range(0,100):
15     s=s+play()/100
16 print(s)

```

- ☐ Simulates a game play 100 times. In each play, a coin is tossed 100 times and player is given money equal to the number of heads he get. The code displays the average money earned by the player amongst all 100 plays.

☐ Anagrams - Part 01 (unit? unit=142&lesson=156)

☐ Anagrams - Part 02 (unit? unit=142&lesson=157)

☐ Anagrams - Part 03 (unit? unit=142&lesson=158)

☐ Facebook Sentiment Analysis (unit? unit=142&lesson=159)

☐ Facebook Sentiment Analysis - Part 01 (unit? unit=142&lesson=160)

☐ Facebook Sentiment Analysis - Part 02 (unit? unit=142&lesson=161)

☐ Facebook Sentiment Analysis - Part 03 (unit? unit=142&lesson=162)

☒ Facebook Sentiment Analysis - Part 04 (unit? unit=142&lesson=163)

☐ Quiz : Assignment 8 (assessment? name=284)

☐ Programming Assignment - 1: Duplicate Elements (/noc20_cs35/progassignment? name=307)

☐ Programming Assignment-2: Panagrams (/noc20_cs35/progassignment? name=308)

☐ Programming Assignment-3: Vowels (/noc20_cs35/progassignment? name=309)

☐ Week 8 Feedback (unit?)

☐ Simulates a game play 100 times. In each play, a coin is tossed 100 times and player is given money equal to the number of heads he get. The code displays the total money earned by the player amongst all 100 plays.

☐ Simulates a game play 100 times. In each play, a coin is tossed 100 times and player is given money equal to the number of heads he get. The code displays the money earned by the player in first play.

☐ none of the above

No, the answer is incorrect.

Score: 0

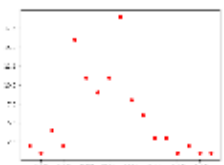
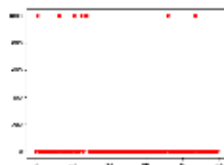
Accepted Answers:

Simulates a game play 100 times. In each play, a coin is tossed 100 times and player is given money equal to the number of heads he get. The code displays the average money earned by the player amongst all 100 plays.

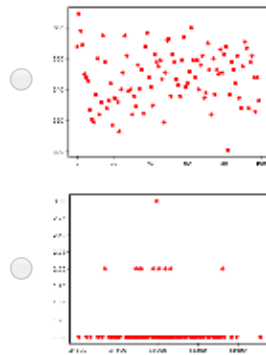
5) Which of the plots in the options is most likely to be generated from the following code?

1 point

```
1 import random
2 import matplotlib.pyplot as plt
3
4 def play():
5     amt=0
6     for i in range(0,100):
7         r=random.randint(1,1000)
8         if(r!=random.randint(1,1000)):
9             amt=amt
10        else:
11            amt=amt+1000
12    return amt
13
14 l=[]
15 for j in range(0,100):
16     s=0
17     for i in range(0,100):
18         s=s+play()
19     l.append(s)
20 x=[]
21 y=[]
22 for each in list(set(l)):
23     x.append(each)
24     y.append(l.count(each))
25 plt.plot(x,y,'ro')
26 plt.show()
```

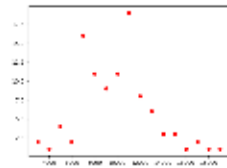


unit=142&lesson=310)

Week 9**Week 10****Week 11****Week 12****Text Transcripts****Download Videos****Books**

No, the answer is incorrect.
Score: 0

Accepted Answers:



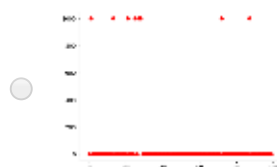
6) Which of the plots in the options is most likely to be generated from the following code?

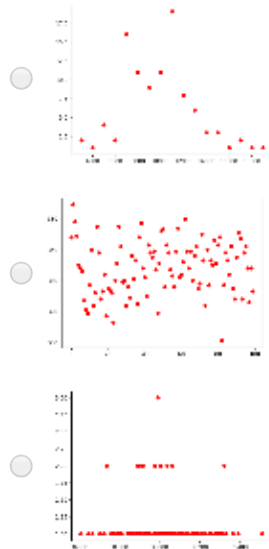
1 point

```

1 import random
2 import matplotlib.pyplot as plt
3
4 def play():
5     amt=0
6     for i in range(0,100):
7         r=random.randint(1,6)
8         amt=amt+r
9     return amt
10
11 l=[]
12 for j in range(0,100):
13     s=0
14     for i in range(0,100):
15         s=s+play()
16     l.append(s)
17 x=[]
18 y=[]
19 for each in list(set(l)):
20     x.append(each)
21     y.append(l.count(each))
22 plt.plot(x,y,'ro')
23 plt.show()
24

```

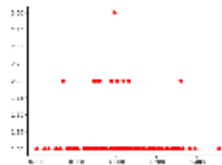




No, the answer is incorrect.

Score: 0

Accepted Answers:



7) What is the output of the following code?

1 point

```
1 dict_age={ }
2 dict_age [ "Arun" ]=20
3 dict_age [ "Bhima" ]=10
4 dict_age [ "Chirag" ]=40
5 dict_age [ "Deepak" ]=30
6
7 dict1=dict_age
8 l=dict_age . values ( )
9 l [0]=90
10 print ( l )
```

- ☐ [20,10,40,30]
- ☐ [90,10,40,30]
- ☐ [10,20,30,40]
- ☐ Error

No, the answer is incorrect.

Score: 0

Accepted Answers:

Error

8) Which of the scenarios in the options does the following code represent?

2 points

```

1 import random
2 dict_age={ }
3 dict_age [ "Arun" ]=20
4 dict_age [ "Bhima" ]=10
5 dict_age [ "Chirag" ]=40
6 dict_age [ "Deepak" ]=30
7
8 l=list ( dict_age . values () )
9
10 dict1={ }
11 l_name=dict_age . keys ()
12 i=0
13 prev=0
14 for each in dict_age :
15     dict1 [ each ]=prev+l [ i ]
16     prev=dict1 [ each ]
17     i=i+1
18 print ( dict1 )
19
20 r=random . randint ( 0 , sum ( dict_age . values () ) )
21 print ( r )
22 for each in dict1 :
23     if ( r<dict1 [ each ] ) :
24         print ( "Give all money to" , each )
25         break

```

- ☐ All money is given to the oldest person
- ☐ All money is given to the youngest person
- ☐ Money is given to a person with a probability proportional to his/her age
- ☐ Money is given to a person with a probability inversely proportional to his/her age

No, the answer is incorrect.

Score: 0

Accepted Answers:

Money is given to a person with a probability proportional to his/her age

9) Which of the scenarios in the options does the following code represent?

1 point

```

1 import random
2 import operator
3
4
5 dict_age={ }
6 dict_age [ "Arun" ]=20
7 dict_age [ "Bhima" ]=10
8 dict_age [ "Chirag" ]=40
9 dict_age [ "Deepak" ]=30
10
11 print ( "Give all money to" , max ( dict_age . items () , key=operator . itemgetter ( 1 ) ) [ 0 ] )
12 l=list ( dict_age . values () )

```

- ☐ All money is given to the oldest person
- ☐ All money is given to the youngest person
- ☐ Money is given to a person with a probability proportional to his/her age
- ☐ Money is given to a person with a probability inversely proportional to his/her age

No, the answer is incorrect.

Score: 0

Accepted Answers:

All money is given to the oldest person

X


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Unit 11 - Week 9

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

- ☐ Natural Language Processing - Author Stylometry (unit? unit=164&lesson=165)

Assignment 9

The due date for submitting this assignment has passed. **Due on 2020-04-01, 23:59 IST.**
As per our records you have not submitted this assignment.

1) The isalpha() function in NLTK

1 point

- ☐ returns true if all the words in a sentence are composed of alphabetic characters and false otherwise
- ☐ returns true if all the characters in a word are alphabets and false otherwise
- ☐ returns true if all the characters in a word are alphabets or numerics and false otherwise
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

returns true if all the characters in a word are alphabets and false otherwise

2) Predict the output

1 point

```
1 my_para="i am to go to KT in A"
2 print(list(my_para))
```

- ☐ ['i', ' ', 'a', ' ', 'm', ' ', ' ', 't', ' ', 'o', ' ', ' ', 'g', ' ', 'o', ' ', ' ', 't', ' ', 'o', ' ', ' ', 'K', ' ', 'T', ' ', ' ', 'i', ' ', 'n', ' ', ' ', 'A']
- ☐ ['i', 'a', 'm', 't', 'o', 'g', 'o', 't', 'o', 'K', 'T', 'i', 'n', 'A']
- ☐ ['i', 'am', 'to', 'go', 'to', 'KT', 'in', 'A']
- ☐ ['i', ' ', 'am', ' ', 'to', ' ', 'go', ' ', 'to', ' ', 'KT', ' ', 'in', ' ', 'A']

No, the answer is incorrect.

Score: 0

Accepted Answers:

['i', ' ', 'a', ' ', 'm', ' ', ' ', 't', ' ', 'o', ' ', ' ', 'g', ' ', 'o', ' ', ' ', 't', ' ', 'o', ' ', ' ', 'K', ' ', 'T', ' ', ' ', 'i', ' ', 'n', ' ', ' ', 'A']

3) Which of the following is a valid function in NLTK?

1 point

☐ Natural Language Processing - Author Stylometry - Part 01 (unit? unit=164&lesson=166)

☐ Natural Language Processing - Author Stylometry - Part 02 (unit? unit=164&lesson=167)

☐ Natural Language Processing - Author Stylometry - Part 03 (unit? unit=164&lesson=168)

☐ Natural Language Processing - Author Stylometry - Part 04 (unit? unit=164&lesson=169)

☐ Natural Language Processing - Author Stylometry - Part 05 (unit? unit=164&lesson=170)

☐ Natural Language Processing - Author Stylometry - Part 06 (unit? unit=164&lesson=171)

☐ Natural Language Processing - Author Stylometry - Part 07 (unit? unit=164&lesson=172)

☐ Natural Language Processing - Author Stylometry - Part 08 (unit? unit=164&lesson=173)

- ☐ freq_dist()
☐ frequency_distribution()
☐ FreqDist()
☐ freqDist()

No, the answer is incorrect.
 Score: 0

Accepted Answers:
FreqDist()

4) Predict the output

1 point

```
1 import networkx as nx
2 G=nx.gnp_random_graph(100,1)
3 print(nx.is_connected(G))
```

- ☐ True
☐ False
☐ "connected"
☐ can not say

No, the answer is incorrect.
 Score: 0

Accepted Answers:
True

5) Which of the following functions when applied to a graph G in networkx will give you its degree of separation? **1 point**

- ☐ is_connected(G)
☐ order(G)
☐ diameter(G)
☐ None of the above

No, the answer is incorrect.
 Score: 0

Accepted Answers:
None of the above

6) What is the degree of separation of the following network?

1 point



- ☐ 1

☐ Natural Language Processing - Author Stylometry - Part 09 (unit? unit=164&lesson=174)

- ☐ 2
☐ 3
☐ 4

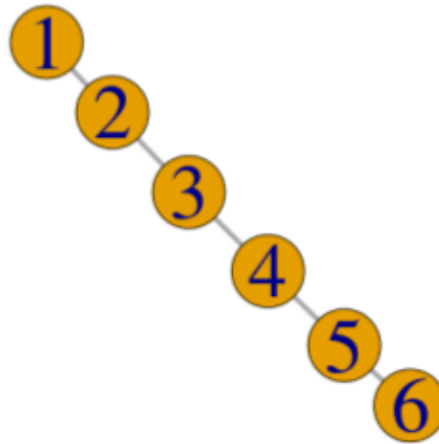
No, the answer is incorrect.
 Score: 0

Accepted Answers:
 1

☐ Natural Language Processing - Author Stylometry - Part 10 (unit? unit=164&lesson=175)

7) What is the degree of separation of the following network?

1 point



☐ Introduction to Networkx - Part 01 (unit? unit=164&lesson=176)

☐ Introduction to Networkx - Part 02 (unit? unit=164&lesson=177)

☐ Six Degrees of Separation : Meet your favourites (unit? unit=164&lesson=178)

- ☐ 1.333
☐ 2
☐ 2.333
☐ 6

No, the answer is incorrect.
 Score: 0

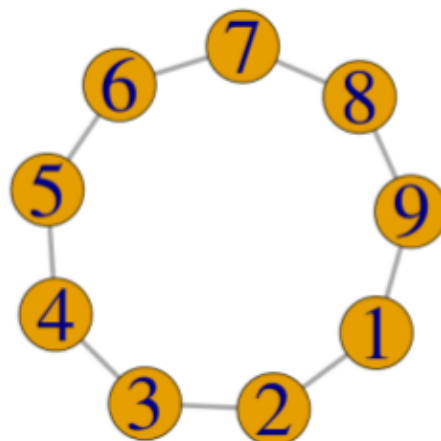
Accepted Answers:
 2.333

☐ Six Degrees of Separation : Meet your favourites - Part 01 (unit? unit=164&lesson=179)

☐ Six Degrees of Separation : Meet your favourites - Part 02 (unit? unit=164&lesson=180)

8) What is the degree of separation of the following network?

1 point



☐ Six Degrees of Separation : Meet your favourites - Part 03 (unit? unit=164&lesson=181)

☐ Area Calculation - Don't Measure (unit? unit=164&lesson=182)

☐ Area Calculation - Don't Measure - Part 01 (unit? unit=164&lesson=183)

- ☐ 1
☐ 2.5
☐ 3.5
☐ 4

- Part 02 (unit?
unit=164&lesson=184)

☐ Area Calculation
- Don't Measure
- Part 03 (unit?
unit=164&lesson=185)

☐ Area Calculation
- Don't Measure
- Part 04 (unit?
unit=164&lesson=186)

☐ Area Calculation
- Don't Measure
- Part 05 (unit?
unit=164&lesson=187)

☐ Area Calculation
- Don't Measure
- Part 06 (unit?
unit=164&lesson=188)

☐ Quiz :
**Assignment 9
(assessment?
name=285)**

☐ Programming
Assignment 1:
Swap the Case
(/noc20_cs35/progassignment?
name=311)

☐ Programming
Assignment-2:
First and Last
(/noc20_cs35/progassignment?
name=312)

☐ Programming
Assignment 3:
Rotate the
matrix
(/noc20_cs35/progassignment?
name=313)

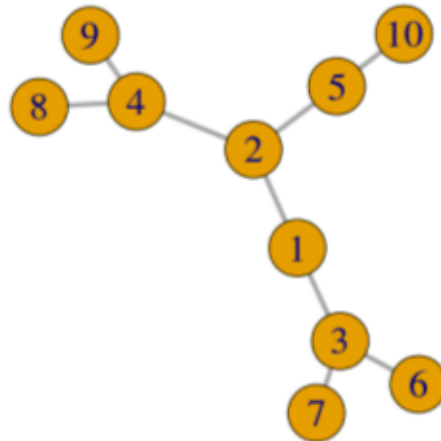
☐ Week 9
Feedback (unit?
unit=164&lesson=314)

No, the answer is incorrect.
Score: 0

Accepted Answers:
2.5

9) What is the degree of separation of the following network?

1 point



☐ 1.82

☐ 2.5

☐ 2.82

☐ 3

No, the answer is incorrect.
Score: 0

Accepted Answers:
2.82

10) Degree of separation of a network is same as its

1 point

☐ Order

☐ Size

☐ Average shortest path length

☐ Number of components

No, the answer is incorrect.
Score: 0

Accepted Answers:
Average shortest path length

Week 10

Week 11

Week 12

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Unit 12 - Week 10

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

- FLAMES - Part 01 (unit? unit=189&lesson=190)

Assignment 10

The due date for submitting this assignment has passed. **Due on 2020-04-08, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Which of the following is a correct representation to convert a string "s" in uppercase in python? **1 point**

- ☐ uppercase(s)
- ☐ s.upper()
- ☐ upper(s)
- ☐ s.uppercase()

No, the answer is incorrect.
Score: 0

Accepted Answers:
s.upper()

2) Predict the output **1 point**

```
1 import string
2 s="Hello"
3 s=s.replace('l','s')
4 print(s)
```

- ☐ Hello
- ☐ Heslo
- ☐ Hesso
- ☐ Heso

No, the answer is incorrect.
Score: 0

Accepted Answers:
Hesso

☐ FLAMES - Part 02 (unit? unit=189&lesson=191)

☐ FLAMES - Part 03 (unit? unit=189&lesson=192)

☐ FLAMES - Part 04 (unit? unit=189&lesson=193)

☐ FLAMES - Part 05 (unit? unit=189&lesson=194)

☐ FLAMES - Part 06 (unit? unit=189&lesson=195)

☐ Data Compression - Part 01 (unit? unit=189&lesson=196)

☐ Data Compression - Part 02 (unit? unit=189&lesson=197)

☐ Data Compression - Part 03 (unit? unit=189&lesson=198)

☐ Data Compression - Part 04 (unit? unit=189&lesson=199)

☐ Data Compression - Part 05 (unit? unit=189&lesson=200)

☐ **Quiz : Assignment 10 (assessment? name=286)**

☐ Programming Assignment-1: Digit (/noc20_cs35/progassignment? name=315)

☐ Programming Assignment-2: Missing Number (/noc20_cs35/progassignment? name=316)

☐ Programming Assignment-3: Rearrangement (/noc20_cs35/progassignment? name=317)

3) Predict the output

```
1 import string
2 s="Hello"
3 s=s.replace("He","s")
4 print(s[2:])
```

- ☐ llo
☐ ello
☐ l
☐ lo

No, the answer is incorrect.
Score: 0

Accepted Answers:
lo

4) Predict the output

```
1 import string
2 s="Hello"
3 print(s.index('ell'))
```

- ☐ [1,2,3]
☐ 1
☐ 2
☐ 3

No, the answer is incorrect.
Score: 0

Accepted Answers:
1

5) Which of the following can be used to see the dimension of a numpy array names 'arr'?

- ☐ dim(arr)
☐ shape(arr)
☐ arr.shape
☐ arr.shape()

No, the answer is incorrect.
Score: 0

Accepted Answers:
arr.shape

6) Predict the output

```
1 import numpy as np
2 a=np.array([1,2,3])
3 b=np.array([[1],[2],[3]])
4 print(a*b)
```

- ☐ [1,4,9]
☐ [2,4,6]
☐ [[123]]

0 points

0 points

1 point

1 point

☐ Week 10
Feedback (unit?
unit=189&lesson=318)

Week 11

Week 12

Text Transcripts

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[246]

[369]]

☐ error

No, the answer is incorrect.

Score: 0

Accepted Answers:

[[123]

[246]

[369]]

7) Predict the output

1 point

```
1 a=[1,2,3]
2 b=[[1],[2],[3]]
3 print(a*b)
```

☐ [1,4,9]

☐ [2,4,6]

☐

[[123]

[246]

[369]]

☐ error

No, the answer is incorrect.

Score: 0

Accepted Answers:

error

8) What is the output of the following code ?

1 point

```
1 example = "snow world"
2 example[3] = 's'
3 print(example)
```

☐ snow

☐ snow world C

☐ Error

☐ snos world

No, the answer is incorrect.

Score: 0

Accepted Answers:

Error

9) What is the output of the following code ?

1 point

```
1 print('ab'.isalpha())
```

☐ True

☐ False

☐ None

☐ Error

No, the answer is incorrect.

Score: 0

Accepted Answers:

True

10) Numpy.array(list), what it does ?

1 point

- ☐ It converts array to list
- ☐ It converts list to array
- ☐ It converts array to array
- ☐ Error

No, the answer is incorrect.

Score: 0

Accepted Answers:

Error

X


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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » **The Joy of Computing using Python (course)**

Announcements (announcements) **About the Course** (https://swayam.gov.in/nd1_noc20_cs35/preview)

Ask a Question (forum) Progress (student/home) Mentor (student/mentor)

Unit 13 - Week 11

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

☒ Browser Automation

Assignment 11

The due date for submitting this assignment has passed. **Due on 2020-04-15, 23:59 IST.**
As per our records you have not submitted this assignment.

1) Which Python library is used for browser automation?

1 point

- ☐ networkx
- ☐ numpy
- ☐ nltk
- ☐ selenium

No, the answer is incorrect.
Score: 0

Accepted Answers:
selenium

2) Which package is used to work with date and time in Python?

1 point

- ☐ timedata
- ☐ datetime
- ☐ dttm
- ☐ tmdt

No, the answer is incorrect.
Score: 0

Accepted Answers:
datetime

3) Which function is used to see the current date and time in Python?

1 point

- ☐ current()
- ☐ today_with_time()
- ☐ now()

Watsapp using Python - Part 01 (unit? unit=201&lesson=202) ☐ Browser Automation Watsapp using Python - Part 02 (unit? unit=201&lesson=203) ☐ Browser Automation Watsapp using Python - Part 03 (unit? unit=201&lesson=204) ☐ Browser Automation Watsapp using Python - Part 04 (unit? unit=201&lesson=205) ☐ Fun with Calendar - Part 01 (unit? unit=201&lesson=206) ☐ Fun with Calendar - Part 02 (unit? unit=201&lesson=207) ☐ Fun with Calendar - Part 03 (unit? unit=201&lesson=208) ☐ Fun with Calendar - Part 04 (unit? unit=201&lesson=209) ☐ Fun with Calendar - Part 05 (unit? unit=201&lesson=210) ☐ Fun with Calendar - Part 06 (unit? unit=201&lesson=211) ☐ Fun with Calendar - Part 07 (unit? unit=201&lesson=212) ☐ Fun with Calendar - Part 08 (unit? unit=201&lesson=213)	☐ none of these No, the answer is incorrect. Score: 0 Accepted Answers: <i>now()</i> 4) Which Python library is used for timezone? 1 point ☐ timezone ☐ python_timezone ☐ pytimezone ☐ pytz No, the answer is incorrect. Score: 0 Accepted Answers: <i>pytz</i> 5) Which of the following statement can be used to see the calendar for January 2020? 1 point ☐ calendar.month(2020, 1) ☐ calendar(2020, 1) ☐ calendar_month(2020, 1) ☐ none of the above No, the answer is incorrect. Score: 0 Accepted Answers: <i>calendar.month(2020, 1)</i> 6) Which statement can be used to come out of an infinite loop? 1 point ☐ continue ☐ break ☐ try ☐ catch No, the answer is incorrect. Score: 0 Accepted Answers: <i>break</i> 7) Which of the following is not a correct conditional block in Python. 1 point ☐ if ☐ else ☐ else_if ☐ elif No, the answer is incorrect. Score: 0 Accepted Answers: <i>else_if</i> 8) The value returned when we use the function isoweekday() is and that for the function weekday() is if the system date is 19th June, 2017 (Monday). 1 point ☐ 0,0 ☐ 0,1
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☐ Fun with Calendar - Part 09 (unit? unit=201&lesson=214)

☐ Fun with Calendar - Part 10 (unit? unit=201&lesson=215)

☐ Fun with Calendar - Part 11 (unit? unit=201&lesson=216)

☐ Fun with Calendar - Part 12 (unit? unit=201&lesson=217)

☐ **Quiz : Assignment 11 (assessment? name=287)**

☐ Programming Assignment-1: Formula (/noc20_cs35/progassignment? name=319)

☐ Programming Assignment-2: word-sorting (/noc20_cs35/progassignment? name=320)

☐ Programming Assignment-3: Numbers (/noc20_cs35/progassignment? name=321)

☐ Week 11 Feedback (unit? unit=201&lesson=322)

Week 12

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☐ 1,0

☐ 1,1

No, the answer is incorrect.
Score: 0

Accepted Answers:
1,0

9) What will be the output of the following Python code if the system date is 18th June, 2017 (Sunday)? **1 point**

```
1 tday=datetime.date.today()
2 print(tday.weekday())
```

☐ 6

☐ 1

☐ 0

☐ 7

No, the answer is incorrect.
Score: 0

Accepted Answers:
6

10) Which of the following functions can be used to find the coordinated universal time, assuming that the datetime module has already been imported? **1 point**

☐ datetime.utcnow()

☐ datetime.datetime.utcnow()

☐ datetime.utcnowow()

☐ datetime.datetime.utcnowow()

No, the answer is incorrect.
Score: 0

Accepted Answers:
datetime.datetime.utcnowow()

X


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Unit 14 - Week 12

Course outline

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Assignment 12

The due date for submitting this assignment has passed. Due on 2020-04-22, 23:59 IST. As per our records you have not submitted this assignment.

1) In web graph, on which we perform Google page ranking

1 point

- ☐ Web pages are the nodes and hyperlinks are the edges
- ☐ Hyperlinks are the nodes and web pages are the edges
- ☐ Both hyperlinks and web pages are nodes
- ☐ Both hyperlinks and web pages are edges

No, the answer is incorrect.

Score: 0

Accepted Answers:

Web pages are the nodes and hyperlinks are the edges

2) In pagerank algorithm on a directed network, we randomly move from a node A to

1 point

- ☐ a random node which is pointing to A
- ☐ a random node which A is pointing to
- ☐ a random node out of all the nodes which are either pointing to A or the nodes which A is pointing to
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

a random node which A is pointing to

3) Which of the following statements is FALSE?

1 point

- ☐ Barbell graph is connected
- ☐ In both complete graph and cycle graph, every node has the same degree

☒ Page Rank -
How does
Google Work ? -
Part 01 (unit?
unit=218&lesson=219)

- ☐ In any star graph having more than 2 nodes, there are at least 2 nodes having the same degree
- ☐ All the above statements are true

No, the answer is incorrect.

Score: 0

Accepted Answers:

All the above statements are true

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How does
Google Work ? -
Part 02 (unit?
unit=218&lesson=220)

4) Which of the following statements choose a random node from a graph G?

1 point

- ☐ random.choice(G)
- ☐ random.node(G.nodes())
- ☐ random.choice(G.nodes())
- ☐ none of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

none of the above

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Part 03 (unit?
unit=218&lesson=221)

☐ Page Rank -
How does
Google Work ? -
Part 04 (unit?
unit=218&lesson=222)

5) The function networkx.pagerank(G) returns

1 point

- ☐ a list
- ☐ a set
- ☐ a dictionary
- ☐ a numpy array

No, the answer is incorrect.

Score: 0

Accepted Answers:

a dictionary

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Part 05 (unit?
unit=218&lesson=223)

☐ Page Rank -
How does
Google Work ? -
Part 06 (unit?
unit=218&lesson=224)

6) Choose one network from following which is not directed.

1 point

- ☐ citation network
- ☐ Follower-followee network of Twitter
- ☐ Supply chain network
- ☐ Friendship network of Facebook

No, the answer is incorrect.

Score: 0

Accepted Answers:

Friendship network of Facebook

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Part 08 (unit?
unit=218&lesson=226)

☐ Page Rank -
How does
Google Work ? -
Part 09 (unit?
unit=218&lesson=227)

7) Which of the following functions is used to make a directed graph?

1 point

- ☐ networkx.DiGraph
- ☐ networkx.digraph
- ☐ networkx.Digraph
- ☐ networkx.diGraph

No, the answer is incorrect.

Score: 0

Accepted Answers:

networkx.DiGraph

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Google Work ? -
Part 10 (unit?
unit=218&lesson=228)

☐ Page Rank -
How does
Google Work ? -
Part 11 (unit?
unit=218&lesson=229)

8) The output of G.out_edges(nodename) is

1 point

- ☐ List of lists
- ☐ List of dictionaries
- ☐ List of vertices

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How does
Google Work ? -
Part 12 (unit?
unit=218&lesson=230)

☐ Page Rank -
How does
Google Work ? -
Part 13 (unit?
unit=218&lesson=231)

☐ Page Rank -
How does
Google Work ? -
Part 14 (unit?
unit=218&lesson=232)

☐ Page Rank -
How does
Google Work ? -
Part 15 (unit?
unit=218&lesson=233)

☐ Page Rank -
How does
Google Work ? -
Part 16 (unit?
unit=218&lesson=234)

☐ Collatz
Conjecture -
Part 01 (unit?
unit=218&lesson=235)

☐ Collatz
Conjecture -
Part 02 (unit?
unit=218&lesson=236)

☐ JOC Conclusion
(unit?
unit=218&lesson=237)

☐ **Quiz :**
Assignment 12
(assessment?
name=288)

☐ Programming
Assignments-1:
Sentence
(/noc20_cs35/progassignment?
name=323)

☐ Programming
Assignment-2:
Letters
(/noc20_cs35/progassignment?
name=324)

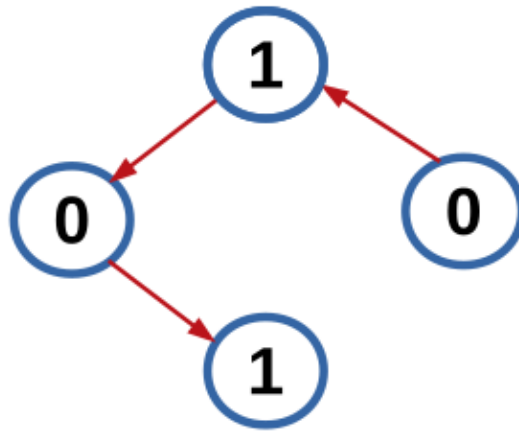
☐ Programming
Assignment-3:
Email ID
(/noc20_cs35/progassignment?
name=325)

☐ List of tuples

No, the answer is incorrect.
Score: 0

Accepted Answers:
List of tuples

9) What happens when a gold coin distribution game is played on the following network? **1 point**



- ☐ One node ends up having all coins and the game stops
- ☐ The game stops after some iterations but one node does not end up having all the coins
- ☐ The game enters an infinite loop
- ☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
One node ends up having all coins and the game stops

10) The number of gold coins in the gold coin distribution game **1 point**

- ☐ keeps increasing in every subsequent iteration
- ☐ keeps decreasing in every subsequent iteration
- ☐ keeps varying randomly in every subsequent iteration
- ☐ remains constant

No, the answer is incorrect.
Score: 0

Accepted Answers:
remains constant

☐ Week 12
Feedback (unit?
unit=218&lesson=326)

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